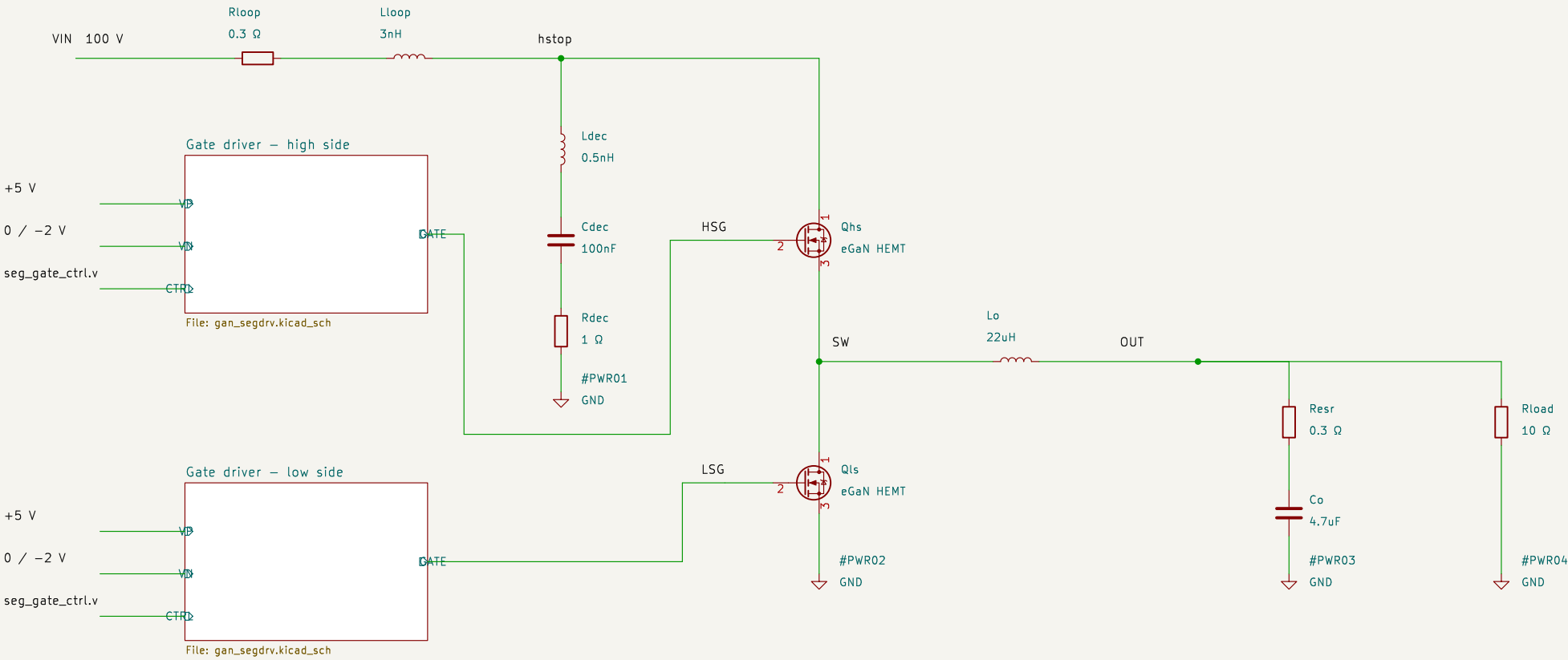


GaN synchronous buck converter – the circuit simulated in sim/buck.cir

100 V in, 48.50 V out at 500 kHz, simulated. Values read from sim/buck.cir and results/buck_sweep.csv at build time.
D = 0.50 alone would give 50.0 V; the 1.50 V short of it is conduction and switching loss.



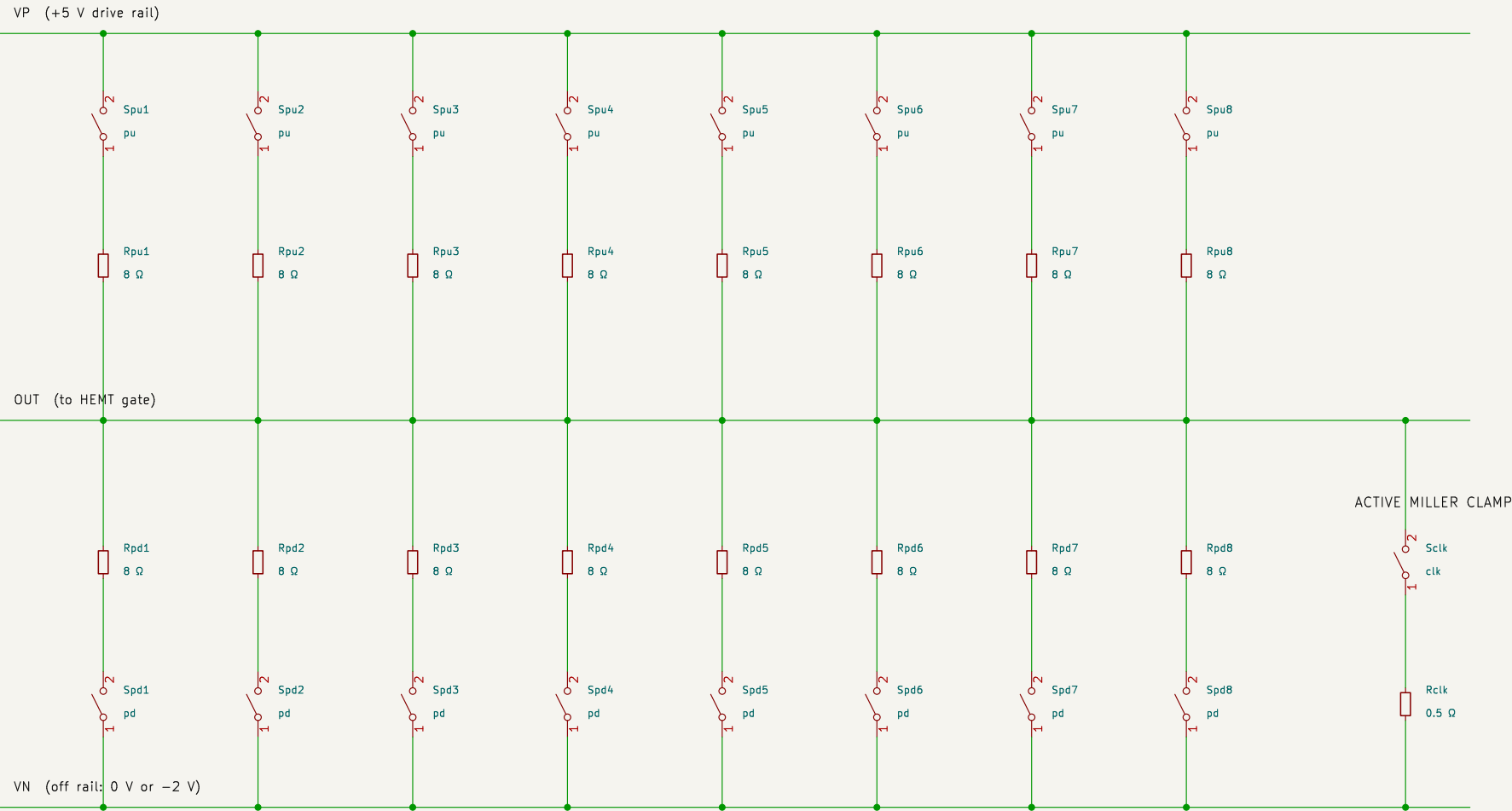
Qhs / Qls are enhancement-mode GaN HEMTs. KiCad has no GaN symbol, so an n-channel enhancement MOSFET symbol is used, as in EPC's own schematics. The device has NO BODY DIODE – reverse conduction during dead time costs $V_{th} + |V_{off}| + I \cdot R_{ds(on)}$, not one diode drop. This sets the dead-time trade.

Ldec / Cdec / Rdec is the bus decoupling. Rdec = 1 ohm is a damping value, not a parasitic: at 20 mOhm this branch rings at ~22 MHz and drove the low-side gate to +11.8 V against a 1.4 V threshold at a 200 V bus.

HSG / LSG are driven by the two hierarchical sheets on the left: the same gan_segdrv.kicad_sch, placed twice. Descend into either to see them.

Not drawn: Vsin, Vshs, Vsls, Vsout are 0 V sources in the netlist used only to sense current. They are measurement points, not components.

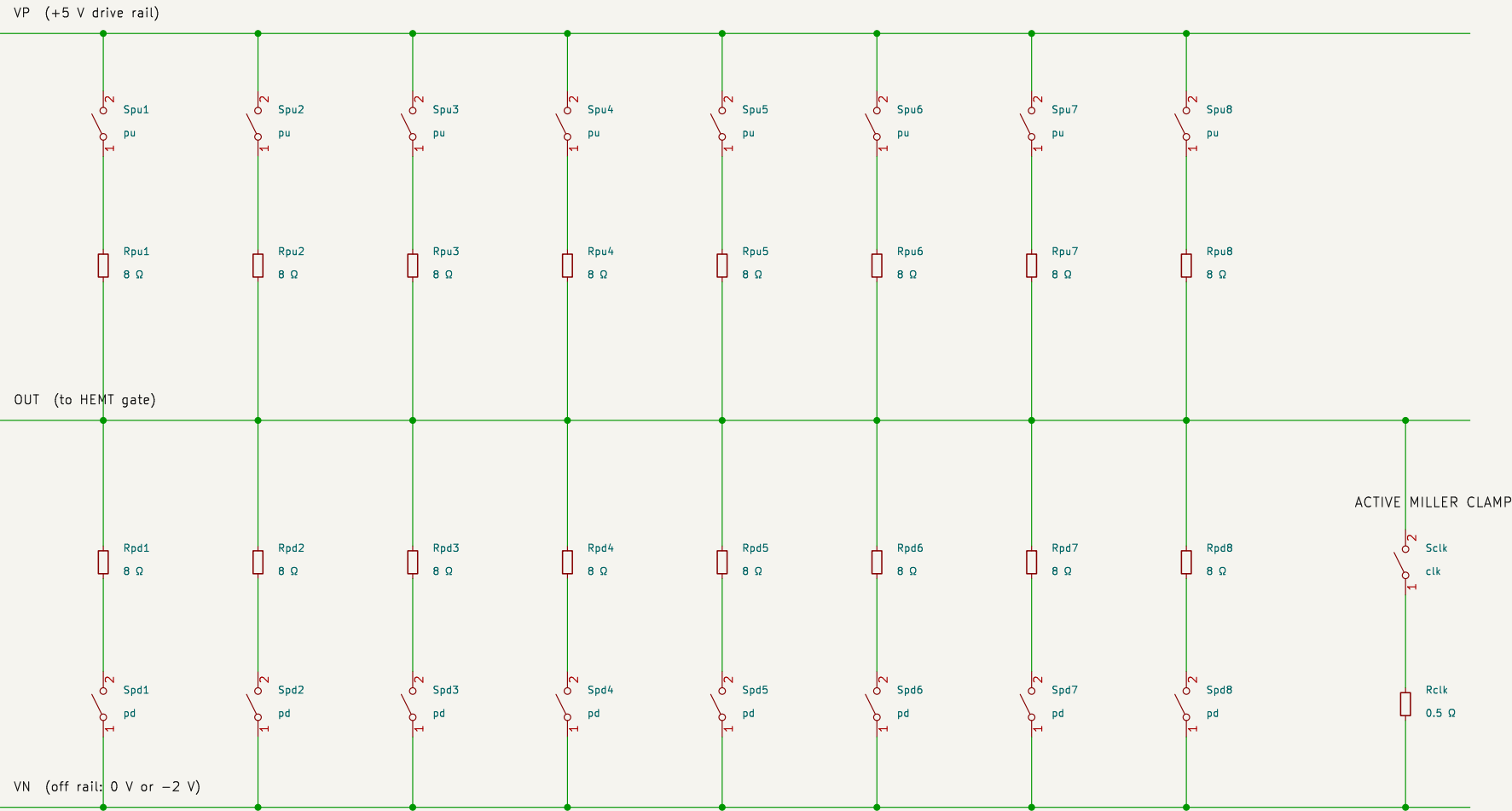
Segmented gate driver – one side of the half–bridge, as simulated
models/segdrv.lib. 8 pull–up slices, 8 pull–down, clamp, selectable off rail.



EACH SLICE IS A SWITCH AND A RESISTOR, AND THE RESISTOR CARRIES THE CODE.
In the netlist slice 3 of the pull–up bank reads:
Rpu3 nu3 out {runit + (npu>=3 ? 0 : 1e9)}
Ask for three or more slices and Rpu3 is runit, 8 ohm, in circuit. Ask for fewer and it becomes 1 Gohm, which is that slice switched out. Drive strength is therefore how many of the eight parallel 8 ohm paths are live: 8 ohm at npu=1, down to 1 ohm at npu=8. The switches are voltage–controlled (.model SWP, Ron 0.01, Roff 1e9) – a behavioural stand–in for the real output transistors, which is stated here rather than implied.

WHAT IS OURS RATHER THAN THE BASE PAPER’S. Sclk and Rclk on the right are the active Miller clamp: it pulls the gate to the off rail through 0.5 ohm whenever the device should be off. VN is selectable to –2 V. models/zhangdrv.lib, the base paper built in the same testbench, has neither; its file says so in as many words. Of the crosstalk margin the clamp is worth +0.82 V and the negative rail a further +2.01 V. The eight slices – the part everyone builds – cost 0.00 % to freeze.

Segmented gate driver – one side of the half–bridge, as simulated
models/segdrv.lib. 8 pull–up slices, 8 pull–down, clamp, selectable off rail.



EACH SLICE IS A SWITCH AND A RESISTOR, AND THE RESISTOR CARRIES THE CODE.
In the netlist slice 3 of the pull–up bank reads:
Rpu3 nu3 out {runit + (npu>=3 ? 0 : 1e9)}
Ask for three or more slices and Rpu3 is runit, 8 ohm, in circuit. Ask for fewer and it becomes 1 Gohm, which is that slice switched out. Drive strength is therefore how many of the eight parallel 8 ohm paths are live: 8 ohm at npu=1, down to 1 ohm at npu=8. The switches are voltage–controlled (.model SWP, Ron 0.01, Roff 1e9) – a behavioural stand–in for the real output transistors, which is stated here rather than implied.

WHAT IS OURS RATHER THAN THE BASE PAPER’S. Sclk and Rclk on the right are the active Miller clamp: it pulls the gate to the off rail through 0.5 ohm whenever the device should be off. VN is selectable to –2 V. models/zhangdrv.lib, the base paper built in the same testbench, has neither; its file says so in as many words. Of the crosstalk margin the clamp is worth +0.82 V and the negative rail a further +2.01 V. The eight slices – the part everyone builds – cost 0.00 % to freeze.